

Gittaveni Sidhartha

ML Research Engineer | LLM & Deep Learning Systems | Model Optimization (Quantization, Pruning) | MLOps & Distributed Training | Python, PyTorch, Hugging Face

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Summary

ML Research Engineer with 5 years of experience implementing and shipping deep learning systems that bridge research prototypes and production deployments across LLM, RAG, and agentic AI stacks. Hands-on track record implementing research algorithms, optimizing models through quantization and efficient inference, and integrating novel methods with infrastructure via MLOps pipelines on Azure, AWS, and GCP. Collaborative researcher-engineer who pairs with applied scientists and product teams to translate novel methods into reliable, reproducible, monitored systems serving hundreds of concurrent users.

Professional Experience

AstraZeneca

Mar 2025 – Present

AI Engineer / DE, USA

- Lexi — Agentic RAG Assistant for Regulated Documents: Architected an end-to-end RAG system delivering accurate, traceable answers from large internal regulated document repositories (SOPs, work instructions, policies), replacing unreliable keyword-based search across the organization.
- Engineered a multi-signal hybrid retrieval pipeline (fuzzy matching, BPE token-based search, vector similarity) that reliably resolves document IDs, partial filenames, and natural-language queries.
- Built a hierarchical document preprocessing framework converting raw files into structured, section-chunked markdown for precise context injection into LLMs.
- Implemented deterministic section injection, resolving LLM-generated reference tags to exact source text at runtime for 100% response grounding and full audit traceability.
- Deployed a scalable, container-based architecture on Azure — Azure OpenAI, Azure AI Search, Blob Storage, and LangGraph orchestration — supporting hundreds of concurrent users.
- Impact: Reduced regulatory document search time by 90%, projected to save 3,750 hours/week at full scale with zero compliance violations and a 30–40% reduction in SME interruption overhead.
- HCP Summarization & Insights with LLMs: Evaluated LLaMA, Phi-3, and Claude 3.7 Sonnet for HCP-level summarization across commercial, behavioral, and publication data; shipped a production insights dashboard generating actionable summaries for sales teams.
- Data Engineering for Model Deployment: Built production pipelines automating data processing and model deployment on AWS, integrated with BI tools for real-time decision-making and reporting.
- Impact: Sharpened understanding of HCP behavior and engagement drivers, enabling more effective field strategies and targeting.
- HCP Early Adoption Modeling (Saphnelo): Designed adoption propensity models (clustering, classification, regression) by analyzing competitor prescribing patterns and forecasting SubQ uptake. Developed production-ready data pipelines using Spark and Python for real-time data processing and model inference.

Freedom Mortgage

Sep 2024 – Feb 2025

Data Scientist, Generative AI (RAG Chatbot for Internal Business Teams) / NJ, USA

- Architected and deployed an enterprise Generative AI RAG chatbot for the Mortgage & Loan Portfolio department using Snowflake Cortex and Claude Sonnet 3.7.
- Built an automated end-to-end pipeline ingesting 720+ unstructured documents from the Confluence API into Snowflake, using Weaviate for vector storage and semantic retrieval and Gemini Flash 2.5 for high-quality, context-aware responses.
- Impact: Delivered a 95% reduction in resolution time with an estimated \$5.28M in annual productivity gains.

Stealth AI Startup (VoiceLink — Voice AI Automation Platform)

May 2024 – Sep 2024

AI Product Engineer / London, UK

- Architected and shipped a production-grade voice AI platform for restaurants deployed on GCP, integrating Deepgram (ASR), Claude/Gemini Vertex AI (LLM), and ElevenLabs/Gemini/Chirp 3 (TTS) to automate inbound/outbound calls — achieving sub-2-second response latency and a 95% call success rate.
- Implemented RAG using Vertex AI over restaurant-specific data for accurate order handling and menu understanding, with full POS system integration for end-to-end automation.
- Impact: Eliminated human dependency for restaurant order management, saving an estimated \$19,200/year per location — a model scalable across enterprise restaurant chains for millions in potential annual savings.

John Deere Financial (NBFC)

Sep 2020 – Aug 2022

Data Scientist (Self-Pay Collections Optimization) / Pune, India

- Led the Self-Pay collection optimization project, developing risk/channel propensity models to assign customers to lower-cost channels and improve collection efficiency.

- Conducted Design of Experiments (DOEs) to enhance pre-field operations, driving process improvements and optimizing CBC charge collections.
- Impact: Achieved a 70–80% improvement in collections and early payments, reduced cost-to-collect, and grew digital repayment retention by 22% while optimizing direct-to-bank settlement cycles.

Samsung R&D

Dec 2019 – Aug 2020

AI/ML Intern / Bengaluru, India

- Bixby Intelligent Capsules (voice-AI): Improved Bixby Intelligent Capsules for expense tracking and crime data analysis, increasing user retention by 22% by implementing NLP and reinforced learning based on 30+ user interviews.

Wawasan Pvt Ltd

May 2019 – Jul 2019

Java Backend Intern / Hyderabad, India

- Developed backend modules for an e-commerce platform using Spring Boot, REST APIs, and MySQL.

Selected Projects

Cognizant (via Forage)

Engaged on end-to-end ML lifecycle, including EDA, predictive modeling, and Machine Learning Production (MLOps) to deliver actionable business insights.

Personal Autonomous Agent ("Jerry")

Self-directed agentic orchestration system in n8n implementing planning, tool-calling, and task routing across personal workflows — a research-style prototype translating multi-agent methods into a running automation stack.

AIG's Summit (Co-Founder)

Built a 200+ member community running weekly live sessions on AI tools and automation for builders and enthusiasts.

Technical Skills

Generative AI & Agentic Systems: LLM Fine-tuning (LoRA, QLoRA, PEFT), RAG, Agentic RAG, Hybrid Search, Vector Similarity, LangChain, LangGraph, AutoGen, A2A, MCP, OpenAI, LLaMA, Phi-3, Claude, Gemini, BERT, Transformers, Attention

Machine Learning: Decision Trees, Ensemble Learning (XGBoost, Boosting), SVM, Clustering, RNN, LSTM, NLP, Reinforcement Learning, Credit Risk Modeling, Propensity Modeling, Sentiment Analysis, Anomaly Detection, Time Series

Frameworks & Libraries: TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers, NLTK, spaCy, Gensim

Cloud & MLOps: AWS (S3, EC2, SageMaker, Bedrock), Azure, GCP, Snowflake, Databricks, Docker, Kubernetes, Git, GitHub

Programming Languages: Python, TypeScript, Java, SQL, R

Data Engineering: Hadoop, Apache Spark, Kafka, Hive, Apache Airflow, Azure Data Factory, Neo4j

Data Science Tools: Pandas, NumPy, SciPy, OpenCV, Matplotlib, Seaborn, Power BI

AI Productivity Tools: Cursor AI, Claude Code, Antigravity, n8n, Bolt.new, Fireflies.ai, Whisper, OpenClaw, Hermes

Education

Indiana Wesleyan University, Marion, IN

Jan 2023 – Aug 2024

Master of Science in Information Technology and Project Management

VNR Vignana Jyothi Institute of Engineering and Technology, India

Jun 2016 – Jun 2020

Bachelor of Technology in Civil & IT Engineering